Task Difficulty and EEG Alpha Asymmetry:
An Amplitude and Frequency Analysis

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Key Words. Task difficulty • EEG alpha asymmetry • Anxiety • Cognitive strategy • Vigilance • Alpha amplitude • Alpha frequency

Abstract. The effects of mathematical, spatial and verbal task difficulty on EEG alpha amplitude and mean frequency asymmetry were investigated. Twenty right-handed subjects with no familial left-handedness (10 female, 10 male) were presented 3 levels of difficulty for each type of task. Difficulty was varied through increasing the rate of auditorily presented numerical stimuli. Also examined were EEG alpha correlates with measures of performance anxiety, subjective difficulty, loss of vigilance, confusion, the tendency to rely on a guessing strategy and perfor-mance. While increasing task difficulty led to right-parietal and posttemporal alpha acceleration for all tasks, task-dependent bilateral changes in alpha frequency were also observed. Increased mathematical task difficulty widened parietal amplitude asymmetry differences between high- and low-performance subjects, and produced performance-dependent changes in left-parietal and right-temporal alpha frequency. A curvilinear relationship between spatial-task difficulty and relative right-hemisphere alpha attenuation was found for the high-performance group only. Finally, numerous correlations were found between alpha measures and subjective and performance variables. Most of these correlations were found to be both task- and difficulty-level-specific. Task anxiety appeared to play a significant role in the determination of parietal- and temporal-lobe asymmetry.

Early investigators of the EEG correlates of mental activity found that in comparison to resting baseline val-ues, cognitive task demands tended to reduce the amount of amplitude of EEG alpha activity (8-13 Hz) [1-3], and to accelerate the dominant frequency within the alpha range [4-6]. While these early electroencepha-lographers interpreted these shifts in alpha as reflecting an increase in mental effort, attention, or the use of visual imagery, subsequent research suggested that ocu-larmotor demands [7, 8] and the timing of saccadic eye movements [9, 10] played a predominant role in determining these parameters of alpha.

However, by the late 1970s, it became apparent that alpha activity could not be conceptualized as a unitary phenomenon, for no single theory could account for its variance across electrode sites, conditions or individuals [11-14]. At about the same time, alpha activity was adopted as a dependent measure to facilitate attempts to functionally localize complex mental processes. More specifically, it was determined that when ratios of homologous alpha amplitude were constructed from recordings of hemisphere-specific task performance, greater relative alpha attenuation was found in the theoretically engaged hemisphere [15]. Recently, similar asymmetries have been observed with respect to patterns of alpha frequency acceleration [16], suggesting that both measures reflect the utilization of distinct cortical regions and their associated cognitive modes. Indeed, the utility of the alpha amplitude asymmetry measures in predicting strategy preference [17, 18], hemisphere-spe-cific task performance [19-22] and emotionality [23, 24], appears well established.

While some progress has been made in elucidating the general cognitive factors that affect alpha activity, the specific performance-related factors that are associated with difference in alpha both within and across individuals remain obscure. No research, for example, has

focused specifically on performance anxiety as a significant source of variance despite numerous reports linking changes in alpha with anxiety [25-28] and suggesting cerebral lateralization of this emotion [29-31]. Similar-ly, while the employment of attention and mental effort have been posited as important variables in understanding the nature of alpha activity [31-35], a study of the associations-guessing strategies and loss of vigilance may have with alpha, in the context of cognitive task perfor-mance, is lacking. Finally, even when some effort has been made to explore specific performance-related factors that influence alpha activity, the results have been inconclusive. For example, some investigators of the effects of task difficulty have suggested that alpha asymmetry increases as a function of difficulty [36, 37]. Other reports have indicated that alpha reflects practice effects [38-40] and that subjective difficulty [41] and performance level [22] may serve as moderating variables. Still others, however, have reported no discernable effect of difficulty on alpha [42-44] or alpha asymmetry [45, 46].

In the present study, an attempt was made to further pinpoint the specific cognitive factors that may influence alpha amplitude and frequency asymmetries. A split-plot design was chosen to assess difficulty effects across ver-bal, spatial and mathematical conditions and to identify the role individual differences in performance may play.

In addition, across-subject correlational analysis was carried out to determine what if any relationship alpha parameters have with self-reported performance anxiety, confusion, subjective difficulty and the tendencies to utilize a guessing strategy or to shift attention away from the task.

Method

Subjects

Subjects were 20 right-handed males (n = 10) and females (n = 10) selected from a small New England liberal-arts college. Selection was based on several factors: initial response to an advertisement self-reported right-handedness according to Edinburgh Handedness Questionnaire [47); no familial history of left-handedness and more than 50% alpha activity in an eye-closed baseline condition. All subjects were in good standing academically, although 2 male subjects and 1 female subject reported a history of learning disabili-ties.

Apparatus

The EEG was recorded from the left-parietal (Ps), right-parietal (Pa), left-posttemporal (Ts) and right-posttemporal (T6) sites and referenced to vertex C, via an ECI Electo-cap of medium size. Electrode positions were selected based on clinical evidence suggesting involvement of these areas in verbal-visual processing [48]. Eye movements were monitored by placing eicetrodes over the outer canti of the eyes. No electrode placement was utilized unless impedance was below 3 kS2.

The EEG and EOG were recorded and amplified by Med Associates AN100 preamplifier/physiological amplifiers set at a 10,000 gain level. The BO amphitier output was then passed to an ANL.

142 EMG/Integrator with the time constant set at 0. The amplified EEG signals were passed first through ANL-142 ECG/EEG pass band filters set at 0.15-150 Hz and 0.4-40 Hz respectively, and subsequently through EEG-500 4-pole Butterworth alpha filters (8-13 Hz). The filtered BEG signals were then either integrated using ANL-160 integrators (time constant - 0.4 s) and analog to digitally converted, or passed directly to the ADC for the purposes of period analysis. Once the signals from all channels were digitized at 100 samples/s, they were passed to a Boltec (IBM-PC clone) computer for further analysis.

Integrated alpha amplitude was computed every 1s as was the integrated eye movement signal. The output of the alpha filters was digitized and the number of zero crossings was counted every 5 s.
These values were then averaged over each task to yield a mean integrated alpha amplitude and mean alpha frequency for each of the EEG channels. Epochs containing eye movement artifacts and
the first enoch (Ist s of the recording) values were eliminated before grand task averaging. Calibration of amplifier, filter and integrator output was carried out before running each subject by passing a 100-MV sinusoidal signal at 10 Hz through each channel in order to calculate a calibration factor used to equate each channel. In addition, after the hirst 10 subjects were run, the amplitier, filter and integrator channels were switched so that hemispheric differences could not be the result of equipment differences. Push button responses by the subject were recorded with DIG 170 Input port, and the time of response was also recorded to determine the number of correct responses.
Materials
Three different types of tasks were presented to subjects, a verbal image-matching task condition, a spatial part-to-whole matching task condition and a serial subtraction task condition. Within each task condition, 3 levels of difficulty were presented by increasing the rate of auditorily presented single-number digits. This was carried out through the use of a variable pitch Radio Shack tape recorder which was situated 2½ feet directly in front of the subject. Within each level of difficulty across all conditions, the stimuli were identi-cal. This was to ensure that differences found across tasks could not be the result of stimulus differences. In addition, pitch was controlled across difficulty levels to reduce potential nondifficulty-
relaico mierterence entects.

All of the 9 tasks were 26 s in length. In the easy task for all conditions, 6 numbers were presented, each number equally spaced in time with an interstimulus interval of 4 s. Nine numbers were presented in the task of intermediate difficulty and in the most difficult task, 12 were presented. The more difficult task stimuli were created on tape by increasing the speed of the original 4 s/stimulus recordings by a factor of 1½ and 2 respectively.
The number of expected correct button press responses varied across conditions and levels of difficulty. For the verbal task condi-tion, 1, 2 and 3 correct button presses characterized the easy, medium and hard tasks respectively. For the spatial tasks, 2, 2 and 3 correct presses characterized the different levels of difficulty and for the subtraction condition, 2, 4 and 6 button presses were expected for tasks in ascending order of difficulty. Button press responses were considered correct only if they occurred at appropriate times during each task. Other button press responses were ignored in the
assessmch or scriormance.
Procedures
Subjects were initially familiarized with the EEG apparatus and briefed as to the general nature of the experiment. They were then asked to nil out s questionnaires, the Mat Anxiety Kaung scale
[49], the Verbalizer-Visualizer Questionnaire [50] and the Fear of Negative Evaluation Scale [51]. Upon completion of these question-
naires, electrodes were attached and detailed instructions concerning the 9 cognitive tasks to be performed were given.

Instructions for the verbal tasks were: Please listen to the following numbers and focus on the sound of each number as vou hear it.
If you hear 2 successive numbers which have the same sound embedded in them, push the button. For example, if you were to hear the number § and then the number 9, you would push the button because they share a long 'i sound. Similarly, if the next number presented was 1 you would press the button again, because 9 and l share an n sound. Remember, compare the sound of the number you hear to the sound of the last number you heard. OK, are there any questions?
Instructions for the spatial tasks were: Please listen to the following numbers and imagine in your mind's eye each number as you near it. Fush the button every time you observes curved nigures in succession. Curved figures are numbers which have a curve in them such as the numbers 3 and 5. Make sure you imagine each number visually. OK, are there any questions?
Instructions for the mathematical tasks were: Please listen to the following numbers and subtract them successively starting from the number 100. If vou miss a number or lose track of a subtotal. please subtract the next number you hear from the last subtotal that you remember. Also, every time your subtraction goes below a 10's place, please indicate such by pushing the button but only after you have completed the subtraction. OK, are there any questions?
Subjects were also instructed as to the use of the pressure-sensi-tive button so that muscle artifacts could be held to a minimum.
One half of the subjects responded with the right hand and one half with the left. The tasks were grouped into 3 conditions, math, verbal and spatial, and each condition was presented in random order.
Tasks varying in difficulty were grouped according to type of task to ensure a proper set before each type of task. Within each condition, the order of task presentation was random, and before each task appropriate instructions were given or repeated.
Upon completion of each task, subjects were asked to fill out a subject's experience questionnaire assessing their estimate of perfor-
mance, degree of performance anxiety, tendency to use a guessing strategy, degree of confusion experienced, tendency to shift attention away from the task, and degree of difficulty encountered. After the first task was completed, subjects were assisted in filling out the first few questions on this questionnaire to facilitate accuracy and ensure questionnaire completion. Upon completing the last ques-tionnaire, electrodes were detached and the subjects thoroughly debriefed.
Artifact Detection System
Epochs containing eye movement artifacts were arbitrarily defined as those 1-second epochs which contained an integrated EOG amplitude value 1.5 standard deviations above the mean. On aver-age, subjects produced 4.27 artifact epochs/tasks with the across-subject range equal to 0-8. For the mean alpha frequency measure which had 5-second epochs, an arbitrary criterion of 2 or more arti-
lact-containing epochs (second epochs) denned an armact-con-taining S-second epoch.
Two subjects were eliminated from the experiment because of excessive eye movement and muscle ten-
SIOn
Results
Analyses of Variances
Difficulty. Difficulty ratings were entered into 3 (task)

X 3 (rate of stimulus presentation) analysis of variance in order to validate that a sufficient range of difficulty was utilized. A significant main effect for difficulty was detected ($F_{2,38} = 12.90$, $p < 0.001$), but in addition, a highly significant effect of task ($F_{2,38} = 25.61$, $p < 0.0001$) and a task X difficulty interaction were found ($F_{4,76} = 3.06$, $p < 0.03$). The most difficult task proved to be the math condition (* = 7.08), the next most difficult, the verbal condition (x = 5.03) and the least difficult, the spatial condition (x = 2.87). Examination of the interaction also indicated that while the levels of difficulty were distinct for the math and verbal conditions ($p < 0.05$), no significant difference was found between the high presentation rate-spatial task and the task of medium presentation rate. Table 1 presents these results.

Amplitude Analyses. When alpha amplitude values were converted to a measure of relative hemisphere activation (P3-P4/P, + P4 and Ts-T%/Ts + T6) and entered into a 3 (task) X 3 (difficulty level) X 2 (within-hemi-sphere location) ANOVA, no significant main or interaction effects were revealed. When analysis of the raw amplitude data was carried out by using a 3 (task) X 3 (difficulty level) X 2 (hemisphere) X 2 (within-hemi-sphere location) ANOVA, only a significant main effect for hemisphere was detected ($F_{1,19} = 28.89$, $p < 0.0001$). Subjects exhibited greater alpha activity over the left hemisphere (x = 0.213) than the right (x - 0.195).

Math Condition. In order to determine if performance level interacted with possible effects of difficulty on alpha amplitude asymmetry, subjects were categorized as high or low performers on the basis of a median split, and ratio data was then entered into a 2 (perfor-mance level) X 3 (difficulty level) X 2 (within-hemi-sphere location) ANOVA. High-performance subjects exhibited a trend to be more relatively right-hemisphere active than low performers in the temporal lobe and relatively more left-hemisphere active in the parietal re-gion.

In order to explore possible relationships between performance level and difficulty further, the data was reanalyzed after the 8 middle-performing subjects were excluded. In this analysis of these extreme-performance subjects, both the performance level X location ($F_{1,9} = 6.63$, $p < 0.03$) and a performance X location X difficulty level interaction ($F_{2,18} = 4.68$, $p < 0.03$) was observed. Table 2 shows these first- and second-order interactions. Notice that for ratio values collapsed across difficulty levels, high-performing subjects are significantly more right lateralized in the temporal lobe than low-performing subjects and marginally more left later-alized in the parietal region ($p < 0.01$). Notice, also, that while there are no significant within-subject difficulty effects, slight increases in left parietal activation in the high-performance group and right parietal activation in the low-performance group leads to a widening of parietal differences as difficulty increases. This increase in asymmetry differences between the two groups does not appear to characterize the asymmetry differences in the temporal area. Examination of the alpha amplitude values at each electrode site was carried out to determine which electrode positions were responsible for these observed asymmetry shifts. A third-order interaction between performance, difficulty level, hemisphere and within-hemisphere location failed to reach significance ($F_{2,18} = 2.62$, $p < 0.1$), but a reliable performance X hemisphere X location effect was found indicating that ($F_{1,9} = 9.01$), high-performance subjects exhibited less

alpha amplitude in the left-parietal and right-temporal regions than low-performance subjects. Additionally, while high-performance subjects exhibited greater alpha attenuation in the right-temporal region than the left, low-scoring subjects produced lower alpha amplitude in the right-parietal region in comparison to the left ($p <$
0.05) (table 3).

Verbal Condition. No reliable main or interaction effects were detected in the analyses of amplitude ratio data for the verbal task condition. Further examination

Table 1. Mean subjective difficulty ratings for each aprior determined level of difficulty for each type of task

Easy
Medium
Hard
Math
6.15
(2.68)
Verbal
4.55
(2.13)
Spatial
2.50
(1.86)
7.00
(2.34)
S.05
(2.37)
2.94
(2.27)
NS
8.10
(1.70)
5.52
(1.76)
3.14
(1.81)

* $p < 0.05$.

Table 2. Mean alpha amplitude asymmetry ratios for high-per-forming group (n - 6) and low-performing group (n - 5) for each level of mathematical task difficulty

Easy
Medium
Hard
High performers
P, - P4/P3 + P4
- .001
(0.188)
NS
-0.005
(0.199)
NS

(0.170)
Ts - T6/Ts + T6
0.098
(0.049)
NS
0.095
(0.039)
NS
0.100 -
(0.048)
Low performers
P3 - P4/P, + P4
NS
NS
(0.078)
(0.063)
0.177
(0.073)
Ts - T6/Ts + T6
-0.061
(0.116)
NS
- 0.050
(0.130)
NS
-0.050
(0.120)|
*p < 0.06; *p < 0.05.
of the extreme performance levels after exclusion of mid-die-level performers revealed no significant effects as well. Analyses of raw amplitude values also revealed only a significant hemispheric effect (F1,17 - 21.01, p <
0.001).
Spatial Condition. When subjects were categorized by performance level based on a median split, and amplitude ratio, data were analyzed in a 3-way ANOVA, a significant performance X difficulty interaction was revealed (F2,34 = 4.38, p < 0.02). As seen in table 4 high-performance subjects produced greater relatively left-
Table 3. Mean alpha amplitude values averaged across difficulty levels for high performers (n - 6) and low performers (n - 5) for the
matnematical concition
Parietal
left
right
High performers
0.170
(0.069)
0.164
(0.035)
Low performers

0.289
(0.112)
0.211
(0.100)
+p < 0.06; **p < 0.01.
Temporal
left
right
0.199
(0.034)
0.164
(0.028)
0.229
(0.100)
NS
0.248
(0.089)
Table 4. Mean alpha amplitude asymmetry values averaged
across parietal and temporal sites for high-performing (n = 11) and
low-performing subjects (n = 8) during spatial condition tasks
High performers
Low performers
*p < 0.05.
Difficulty level
easy
medium
hard
0.062
(0.054)
0.048
(0.045)
NS
0.050
(0.054)
0.050
(0.043)
NS
0.062
(0.054)
0.044
(0.044)
hemisphere activation during the task when stimuli were presented at an intermediate rate. However, analysis of the amplitude values underlying this effect yielded only slight evidence for a performance X difficulty X hemisphere interaction ($F_{2,34} = 2.80$, $p < 0.1$). Thus both activation of the left hemisphere in some subjects and increases in alpha in right hemisphere in others appear to be responsible for this asymmetry effect.
Frequency Analyses

When the average alpha frequency for each electrode site was converted into ratio scores for the parietal and temporal areas and then entered into 3 (task condition)
X 3 (difficulty level) X 2 (within-hemisphere location)
ANOVA, a significant main effect for location ($F_{1,19} = 107.76$, $p < 0.0001$) and, unexpectedly, a main effect for
difficulty were revealed ($F_{2,38} = 6.54$, $p < 0.004$). The
Earle
parietal region ($x = 0.047$) was found to be relatively
more left hemisphere active than the temporal region
($x = -0.058$). Relatively right-hemispheric alpha acceleration increased as a function of difficulty level (easy X =
-0.002; medium $x = -0.006$; hard $x = -0.009$) with all
possible comparisons yielding a significant difference ($p < 0.05$). Further examination of how alpha frequency varied at each electrode site revealed a significant difficulty X hemisphere interaction ($F_{2,38} = 3.70$, $p < 0.04$).
Figure 1 illustrates that alpha frequency tended to increase in the right hemisphere for all tasks as difficulty increased, the easy task mean being reliably lower than the hard task mean ($p < 0.02$). In addition, while left-hemispheric alpha frequency also tended to decline, no reliable decrease was discovered. Nevertheless, for the hard task, in contrast to lower difficulty levels, right-hemispheric alpha frequency was found to be significantly higher than the left-hemispheric mean value ($p <$
0.02).
A 3-way interaction, task * difficulty level X loca-
tion, also reached significance ($F_{4,76} = 3.45$, $p < 0.02$).
Figure 2a-c demonstrates that while temporal lobe frequency did not vary appreciably across tasks or difficulty levels, parietal frequency varied dramatically across both. For all tasks, parietal frequency was significantly higher than temporal frequency during the easy tasks (math - $p < 0.02$; verbal - $p < 0.02$; spatial - $p < 0.03$).
Similarly, for all tasks, this difference declined at the intermediate level of difficulty. However, for the verbal condition, parietal frequency decreased as the difficulty of the task increased (easy vs. hard: $p < 0.03$), but in the spatial task condition parietal frequency tended to increase as a function of difficulty (easy vs. hard: $p < 0.01$; medium vs. hard: $p < 0.04$). Indeed, during the spatial task of highest stimulus presentation rate, parietal alpha frequencies were again higher than temporal lobe values ($p < 0.01$).
Finally, in this task comparison analysis, a significant main effect for within-hemisphere location was detected
($F_{1,19} = 5.11$, $p < 0.04$). However, a highly significant
location X hemisphere interaction was found as well
($F_{1,19} = 120.40$, $p < 0.0001$). Multiple comparisons indicated that the left-parietal ($x = 9.64$) and the right-tem-
poral ($x = 9.63$) areas produced higher average alpha frequencies than either the right-parietal region ($x = 8.78$)
or the left-parietal area ($x = 8.59$; $p < 0.01$).
Math Condition Alpha frequency ratios were analyzed according to performance (based on median split), difficulty level and within-hemisphere location. A significant location effect was found for all conditions ($F_{1,17} =$

Fig. 1. Mean EEG alpha frequency for left and right hemispheres averaged across parietal and temporal sites and averaged across all tasks for easy, medium and hard task difficulty conditions. Hard:
right hemisphere > left hemisphere, p < 0.05.
Fig. 2.a Mean EEG alpha frequency averaged across hemi-spheres for parietal (x-
-X) and temporal sites (•--) for thel
mathematical condition. Easy: parietal > temporal; p < 0.02.
b Mean EEG alpha frequency averaged across hemisphere for parietal (x—-x) and temporal sites (*--) for the verbal condition. Easy: parietal > temporal; p < 0.02. Parietal: easy > hard; p < 0.03. e Mean EEG alpha frequency averaged across hemispheres for parietal (x—x) and temporal (•
•) sites for the spatial condition.
Easy: parietal > temporal: p < 0.03. Hard: parietal > temporal: p <
0.01. Parietal: easy < medium; p < 0.04; easy < hard; p < 0.01.
98.72, p < 0.0001), but a reliable performance level X dificulty level X location interaction was also detected (F2,34 7.00, p < 0.004). As seen in figure 3, the shifts in asymmetry from the task of intermediate difficulty to the most difficult are primarily responsible for this overall interaction.
While the high-performance subjects
showed a slight trend toward a decrease in parietal (p <
0.07) and temporal asymmetry (p < 0.1), low-perfor-mance subjects exhibited nonsignificant trends for asym-metry to increase (parietal: p < 0.1; temporal: p <
0.1).
An analysis of the average alpha frequency at each level yielded a significant performance X difficulty level
interaction (F2,34 = 8.30, p < 0.002) and a performance
X difficulty X hemisphere X location interaction (F2,34
= 9.57, p < 0.001). Figure 4 illustrates these effects. Note
Fig. 3. Mean EEG alpha frequency asymmetry ratios for parietal
and temporal sites for high performers (n = 9) and low performers
(n = 10) during math condition at each level of difficulty. Medium -
hard differences: p < 0.1.0-
= Mign pertormers partetal, ---
= high performers temporal; x-
-x = low performers parietal;
x---x = low performers temporal.
Fig. 4. Mean EEG alpha frequency for each electrode site for
high performers (n - 9) and low performers (LP; n = 10) during
mathematics condition at each level of difficulty. Easy: high performers > LP; p < 0.04; right temporal (RT) p < 0.01. LP: easy > medium; p < 0.02; RT, p < 0.05; medium < hard, LP, p < 0.01;
RT, p < 0.01. High performers: medium > hard, LP, p < 0.02; RT, p < 0.05. RP - Right parietal.
MEAN
ALPHA
9.0
8.5
EASY

MEDIUM
HARD
that for the least difficult task, high performers produced faster alpha frequencies in the left-parietal ($p < 0.04$) and right-temporal regions ($p < 0.01$). As difficulty increased to the intermediate level, these performance level differences also increased due to a reduction in frequency in both brain areas for low-performance subjects (P3: $p < 0.02$; T6: $p < 0.02$). As difficulty increased fur-ther, however, low-performance subjects manifested an increase in right temporal ($p < 0.01$) and left parietal alpha frequency ($p < 0.01$) while high-performing subjects exhibited a decrease in these regions (P3: $p < 0.02$;
T6: $p < 0.05$). Thus, during the most difficult math task, no reliable differences were found between high and low performers at any electrode site. With regard to the observed asymmetry interaction mentioned previously, it appears that changes in the right temporal and left parietal regions were primarily responsible since difficulty had no significant effects on activity recorded from the other 2 leads.
Verbal Condition. Analysis of frequency asymmetry ratios as a function of performance level (median split), difficulty level and brain lobe yielded no significant main or interaction effects. However, when alpha frequency at each electrode site was analyzed, a number of significant effects were found. A reliable performance X difficulty effect was described above, with alpha frequencies decreasing as a function of difficulty in the parietal region but not the temporal region (fig. 2b). In this analysis, however, a performance X difficulty X
location effect was also observed ($F_{2,34} = 4.42$, $p < 0.02$).
While both groups exhibited a decrease in parietal lobe alpha frequencies as difficulty increased, only low-per-formance subjects exhibited a trend towards an increase in temporal alpha frequencies (easy - hard: $p < 0.06$).
Figure 5 depicts these performance-level-related shift in alpha frequency.
Spatial Condition. No reliable performance-related effects were found for asymmetry or raw alpha frequency
Fig. 5. Mean EEG alpha frequency averaged across hemispheres for parietal and temporal sites for high performers (n - 10) and low
performers (n = 9) during the verbal task at each level of difficulty.
Easy > hard: parietal, $p < 0.05$. Low performers: easy < hard, p <
-x = Low performers parietal, x---x = low performers
* - high performers parietal; • - high perform-
temporal:
ers temporal.
data when EEG values were analyzed for the spatial task.
No further analysis was attempted due to the high proportion of subjects who achieved a perfect score for this condition.
Correlational Analyses
Math Condition - Amplitude Analyses. Math task anxiety, confusion and difficulty were highly intercorre-lated and were associated with relatively right-parietal (anxiety: $r[18] - 0.53$, $p < 0.02$; conclusion: $r[18] -$
0.53, $p < 0.02$; difficulty: $r[18] = 0.57$, $p < 0.01$) and
relative left-temporal alpha amplitude activation (anxi-
ety: $r[18] = 0.59$, $p < 0.01$; confusion: $r[18] = 0.49$, p<
0.05; difficulty: $r[18] = -0.52$, $p < 0.02$). Examination
of raw amplitude values at each electrode site, indicated that for anxiety, differences in the activity of both hemispheres were responsible for these individual differences in asymmetry

since no reliable correlations were found with alpha activity recorded from any electrode site. In contrast, the degree of reported confusion was found to be correlated with right-temporal alpha activity (r [18] = 0.46, p < 0.05) while difficulty ratings increased as a function of left-parietal alpha activity (r [18] = 0.47, p < 0.05). Common to these subjective experience categories was a tendency for ratings-parietal asymmetry relationships to strengthen as a function of within-subject diffi-culty. For example, while state anxiety was not reliably related to parietal asymmetry during the easiest of math conditions (r [18] = 0.42, NS), the magnitude of the correlations increased as the task became more difficult (medium: r [18] = 0.51, p < 0.02; hard: r [18] - 0.61, p< 0.01).

Other important relationships between subjective ratings and EEG amplitude values include attention shift-ing, guessing tendencies, and trait anxiety. The use of a guessing strategy was associated with relatively left temporal amplitude activation (r [18] = 0.49, p < 0.03), apparently tied to differences in both hemispheres across subjects, and was most prominent in the easiest of math task conditions (r [18] = -0.49, p < 0.03). In contrast to the case of state anxiety, trait anxiety was found to be associated with relatively right parietal activation only during the easiest of tasks (r [18] = 0.45, p < 0.05).

Weak trends for the tendency to shift attention away from the tasks were also observed for relatively right parietal (r [18] = 0.43, p < 0.06) and left-temporal amplitude asymmetry (r [18] = 0.43, p < 0.06). Unlike anxiety, confusion and difficulty, however, high loss of vigilance was reliably related to lower alpha amplitude in the right parietal area (r [18] = -0.51, p < 0.03). While the asymmetry correlations only reached significance for the most difficult of math tasks (parietal: r [18] - 0.49, p < 0.03; temporal: r [18] - -0.52, p < 0.02), reliable alpha amplitude-attention switching associations were detected for the right-parietal region during both the easiest (r [18] = - 0.48, p < 0.04) and most difficult tasks (r [18] = - 0.45, p < 0.05).

Math Condition - Frequency Analyses. Althouth most subjective rating-EEG alpha relationships were found in the amplitude domain, a number of notable associations were found in the frequency domain for the mathematics condition. An overall task association between difficulty and relatively right temporal frequency acceleration was found (r [18] = -0.45, p < 0.05) and this was most evident for the most difficult task (r [18] = -0.62, p 0.004). A breakdown of the task ratio by electrode site revealed that subjects who specifically exhibited lower left-temporal alpha frequency experienced greater difficulty (r [18] = -0.51, p < 0.03). In the task of intermediate difficulty, a more global set of relationships were found between difficulty ratings and alpha frequency. High difficulty was reliably associated with lower left-

(r [18] = 0.53, p < 0.02) and marginally with right-pari-etal alpha frequency (r [18] = 0.43, p < 0.06).

Task confusion, similar to difficulty, was related to low left-temporal frequency during the most difficult task (r [18] - 0.49, p < 0.03) and marginally associated

with left- ($r[18] = 0.43$, $p < 0.06$) and right-parietal
alpha frequency ($r[18] = -0.43$, $p < 0.06$) during the
task of medium difficulty.
Also of interest are the observations that while no relative relationships were detected between state anxiety and alpha frequency measures, math trait anxiety was
found to be related to low left-parietal ($r[18] = -0.50$,
$p < 0.03$) and right-temporal frequency ($r[18] = -0.44$, $p < 0.05$) during the task of intermediate difficulty.
Spatial Condition - Amplitude Analyses. For spatial tasks taken as a whole, only state anxiety proved to be reliably associated with amplitude asymmetry. Higher anxiety was observed among subjects exhibiting relatively right-parietal activation ($r[18] = 0.47$, $p < 0.04$),
and this relationship was most prominent during the
easy-task condition ($r[18] = 0.54$, $p < 0.02$). As in the
math condition, differences in each hemisphere appeared to be responsible for the overall task asymmetry-anxiety relationship.
Right-temporal
alpha activity,
however, was found to increase as a function of task anxiety during the easy spatial task ($r[18] = 0.44$, $p < 0.05$).
While few alpha amplitude asymmetry-subjective rating relationships were found, a number of bilateral alpha associations emerged. High difficulty and confusion appeared to be related with high alpha activity to both the
left-temporal (difficulty: $r[18) = 0.47$, $p. < 0.04$; confu-
sion: $r[18] = 0.45$, $p < 0.05$) and right-temporal regions
(difficulty: $r[18] = 0.52$, $p < 0.02$; confusion: $r[18] - 0.50$, $p < 0.03$). Finally, in the spatial task where stimuli were presented at an intermediate rate, subjective difficulty was related to alpha activity in all electrode sites
(P3: $r[18] = 0.52$, $p < 0.02$; P4: $r[18] = 0.46$, $p < 0.04$;
Ts: $r[18] = 0.52$, $p < 0.02$; T6: $r[18] = 0.62$, $p < 0.005$).
The only other amplitude relationship of note for the spatial tasks was a negative correlation between alpha amplitude recorded from the right parietal region and the tendency to shift attention away from the task ($r[18] = 0.46$, $p < 0.05$).
Spatial Condition - Frequency Analyses. Few significant correlations were found between EEG frequency measures and questionnaire data for the spatial task con-ditions. A marginal trend for relatively left parietal frequency activation to be associated with the tendency to Earle
shift attention was found for the spatial tasks ($r[18] = 0.44$, $p < 0.06$). This relationship reached significance
for the least difficult task ($r[18] = 0.52$, $p < 0.02$).
Verbal Condition - Amplitude Analyses. No reliable correlations were found between alpha amplitude asymmetry measures and behavioral/subjective experience questionnaire data for the verbal condition. However, when alpha amplitude relationships were examined, a number of significant and rather dramatic findings emerged. Similar to the other task conditions, the tendency to shift attention away from the verbal task was
associated with right parietal activation ($r[18] = 0.54$,

p < 0.02). This correlation was found to be significant for the easiest (r [18] - 0.49, p < 0.03) and most difficult
(r [18] = -0.45, p < 0.05) tasks. More unexpected, how-
ever, was how across-subject difficulty modified the relationship between confusion and alpha amplitude. Task confusion or uncertainty experienced during the least difficult task was associated with lower right-parietal
alpha amplitude (r [18] = -0.62, p < 0.004). In the most
difficult verbal task, however, confusion among subjects increased as a positive function of right-parietal alpha
(r [18] = 0.48, p < 0.04). Interestingly, a critical difference between the 2 tasks was a significant tendency for anxiety (r [18] - -0.47, p < 0.04), to be associated with the right-parietal activation during the least difficult task but not during the most difficult task.
Verbal Condition - Frequency Analyses. Similar to the amplitude analysis, alpha frequency asymmetry analysis provided little information regarding EEG-subjective experience relationships. An exception was the detection of a significant association between the tendency to guess and relatively left-parietal frequency activation (r [18] - 0.50, p < 0.03) for the task of intermediate difficulty.
Inspection of frequency data at each electrode site also yielded few reliable correlations between EEG measures and experiental phenomena. Math trait anxiety was found to be associated with lower frequency in the left-parietal region for the easiest of verbal task (r [18] =
- 0.49, p < 0.03) and lower frequency in the right-tem-
poral region for the most difficult task (r [18] = -0.52,
p < 0.02).
Performance
Although the analyses of variance detected significant associations between EEG alpha measures and perfor-mance, correlational analyses were carried out in addition to determine if specific relationships might have been obscured due to common error variance and to assess what relationships might exist between the EEG
alpha measures recorded during performance on easier tasks with those on more difficult tasks!. One subject was dropped from the performance analysis because of inconsistent button press. Most performance-EEG alpha measure correlations were found for the mathematics task. Relatively right-temporal amplitude activation was significantly related to the overall math condition performance (r [17] = 0.45, p < 0.05). Left-parietal (r [17) =
-0.53, p < 0.03) and right-temporal (r [17] - - 0.65, p<
0.002) alpha amplitude were also related to overall performance while high mean alpha frequencies in the same regions predicted performance as well (P3: r [17] = 0.59,
p < 0.01; T6: r [17] = 0.44, p < 0.1).
A rather interesting pattern in the performance data was the tendency for EEG recorded during easier tasks to predict more difficult task performance. Relatively right-temporal alpha amplitude activation in the easy task predicted performance during the most difficult task (r [17] = 0.49, p < 0.04) while concurrent measures were not significantly related. Similarly, high left-parietal alpha
frequency recorded during the easy task (r [17] = 0.47,
p < 0.05) and task of medium difficulty (r [17] - 0.68, p < 0.002) were associated with performance on the most difficult task, but difficult task performance and EEG alpha frequency were found to be unrelated. Right-temporal lobe frequency measures showed the

same pattern of relationships while right-parietal alpha frequency measured during the medium difficult task also predicted task performance ($r [17] = 0.47$, $p < 0.05$).
Finally, the only other reliable relationship detected for other conditions was for verbal performance to increase across subjects as a function of decreasing alpha frequency in the right temporal region of the brain during the most difficult verbal task ($r [17] = 0.47$, $p < 0.04$).

Discussion

Despite the observation that manipulation of task difficulty produced few significant effects on alpha amplitude at parietal and posttemporal sites, measurable effects were found when performance level was included as a variable and alpha asymmetry was used as a dependent

' Sex and task interaction for alpha amplitude asymmetry - ($F 2$,
$36 - 5.10$, $p < 0.02$). Males exhibited greater relative right-hemi-sphere activation for the spatial condition in comparison to the verbal condition. Females showed no significant difference between the two types of task.

measure. In the math condition, as difficulty increased, a widening of the parietal asymmetry differences between high and low performances was observed. In the spatial condition, while the manipulation of difficulty was not totally successful, high performers exhibited greater relatively left-hemispheric alpha attenuation during the task in which stimuli were presented at an intermediate rate, while low-performing subjects showed no reliable across-task difficulty differences. Collectively, these data confirm past research in this laboratory [22, 40] and suggest further that for verbal-visual processing in general, across-task difficulty and alpha amplitude asymmetry may be curvilinearly related.

It appears likely that the widening of parietal asymmetry differences between performance groups in the math condition was the result of increasing anxiety and confusion. Both of these variables were correlated with parietal amplitude asymmetry, and as difficulty increased the magnitude of these associations increased as well. However, increasing stress or tension cannot be the only reason why asymmetry may change as a function of difficulty. If this were the case, increases in relatively right parietal activation would have been expected primarily within the low-performance group. Earle [22] found that the widening of parietal asymmetry differences between performance groups in arithmetic tasks of increasing difficulty was produced by high-performing subjects exhibiting greater relatively left-hemisphere activation as difficulty increased. This discrepancy with the findings of past research might be explained by the earlier study's use of an easier set of mathematical tasks, subjects of higher ability and no time requirements for performance.

The U-shaped function observed between stimulus presentation rate and relatively right-hemispheric amplitude activation in the spatial task for high performers was unanticipated. The task required identification of stimuli according to physical characteristics and a part-to-whole matching procedure, both of which are purported to differentially engage the right hemisphere [52,
53]. Moreover, subjects were explicitly instructed to utilize visual imagery, a cognitive strategy that at least some authors have assumed to be right-hemisphere-specific [54, 55]. In retrospect, however, the stimuli were most probably verbally encoded as well. The task also required close attention to detail and the use of higher-order processes such as categorization and short-term memory. Since utilization of these cognitive processes is associated with activation of the left-hemisphere in tasks involving spatial analysis [56-60], one could alter-
natively have expected an increase presentation rate to increase left- instead of right-hemisphere involvement.

While this task reclassification appears adequate as an explanation for the initial difficulty effects, it fails to account for the observed increase in relative right-hemi-sphere asymmetry as stimulus presentation rate increased further or why both effects were found only for the high-performance group. Conceivably, as the stimulus presentation rate increased to its maximum, high performers may have switched from a verbally mediated-feature-oriented strategy to a more nonverbal 'holistic' method of solution. Such a shift in strategy has been hypothesized to underlie difficulty-related changes in hemifield advantage in reaction time studies of spatial task performance [61, 62]. However, difficulty-induced changes in strategy would most logically be exhibited by low- and not high-performing subjects. Moreover, since no assessment of specific strategy type was included in this investigation, and because sex differences are likely to have influenced asymmetry in this task, further study is necessary to determine the causes of the observed asymmetry differences between groups and across difficulty levels.

If alpha amplitude can be categorized as not particularly responsive to increases in task difficulty, alpha frequency might be described as being dramatically sensi-tive. The increase in relatively and absolutely right-hemispheric frequency activation across all task is consistent with past research of the effects of performance stress and anxiety. In several lateral eye movement studies increased performance anxiety or social stress has been shown to increase the proportion of left lateral eye [29, 31, 63, but see 64] movements indicative of greater right-hemisphere activation. In our laboratory, we have found that the degree to which normal college studies report an increase in stress from a neutral condition to a stressful one involving cognitive performance,
the
greater the shift in relative left looking they exhibit [63].

However, while 2 reports link EEG asymmetry measures to directional tendencies in lateral eye movements [65, 66], no study has yet investigated a possible link with alpha frequency asymmetry. Moreover, in the current study, across-subject analyses of negative state reports (involving confusion, anxiety, loss of vigilance and difficulty) showed that high alpha frequency, irrespective of hemisphere, was typically associated with less, not more of these dimensions. Thus, it appears that further research is necessary to determine what this diffi-culty-induced right-hemispheric alpha acceleration sig-nifies.

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In addition to the general effect on alpha frequency asymmetry, within-subject difficulty had specific interactive effects with performance level in both the math and verbal conditions. High performers in the math condition showed a decline in both left parietal and right temporal alpha frequency between the task of intermediate difficulty and the most difficult task. Lower-per-forming subjects exhibited a similar decline at easier levels of apriori determined difficulty. Since left-parietal and right-temporal frequency were positively related to performance during the 2 levels of lower difficulty, it is possible that the reduction in frequency in these areas reflects reduced efficiency in processing or overtaxation of the cortical systems dedicated to the task. Low-per-forming subjects also, however, exhibited an increase in frequency in these regions when difficulty increased to the most difficult level. A marginal trend was detected for guessing to be associated with right-temporal alpha frequency during this most difficult task. Perhaps excessive demands on low-ability subjects, at higher levels of difficulty, induced a switch in strategy resulting in an increase in frequency activation but less accurate perfor-mance. This proposed shift in strategy and its effects on alpha frequencies in these cortical regions might explain why frequency measures recorded

during the less difficult task predicted performance on the most difficult task while concurrent measures of frequency and performance for the hardest task were found to be unrelated.
In the verbal condition, while both high- and low-performance groups exhibited a tendency for alpha frequency in the parietal lobe to decrease as difficulty increased, low-scoring subjects tended to show an increase in temporal lobe alpha frequencies. The across-subject correlations were not helpful in deciphering the cause of this interaction, but an ad hoc explanation for the observed overall decrease in parietal alpha frequencies may be offered. Perhaps, as difficulty increased, parietal alpha frequency decreased due to increased short-term memory interference [67] and inability to make rapid judgments in an otherwise simple but unfamiliar perceptual classification task [68]. Limited support for this thesis is the finding that high-performance subjects who exhibited high levels of confusion during the most difficult verbal tasks exhibited relatively lower left-parietal alpha frequency.
In addition to its sensory integrative functions, the parietal region is intimately engaged in the processing of spatial arrangements and relationships. That EEG alpha frequency reflects this specialization is supported by the observation in the present study that increasing the rate

of stimulus presentation in a rather simple spatial-matching task leads to an increase in parietal lobe alpha frequencies.
Parenthetically, high-parietal alpha fre-
quency either in the left or the right hemisphere appears to be associated with spatial strategy preferences. In a recent study of visual imagery, visualizers tended to manifest higher left-parietal alpha frequency than ver-balizers while the tendency to utilize an image-mediated holistic strategy was associated with high right-parietal alpha frequency [75].
Finally, that task difficulty is an important variable to consider in EEG alpha activity studies of cognition is attested to by the numerous effects it has on across-sub-ject correlations between EEG alpha measures and the subjective/behavioral dimensions of cognitive task per-formance. The effects of increasing task difficulty included the strengthening of correlations (e.g. math state anxiety and relatively right-parietal amplitude activa-tion), the weakening of correlations (e.g. math trait anxiety and relatively right-parietal amplitude activation), the shifting of reliable relationships from one brain area to another (e.g. math trait anxiety and left-parietal alpha frequency during the easy math task to right-temporal alpha frequency in the most difficult task) and the reversal of the sign of a correlation (e.g. verbal task confusion and parietal alpha amplitude).
Subjective Difficulty and Performance
Recognizing that differences in the way difficulty is defined may affect conclusions as to how difficulty is related to the EEG (69], and also noting that the relationships between subjective difficulty and EEG alpha may vary according to the level of apriori determined diffi-culty, a summary of these relationships is presented in conjunction with performance data to elucidate which areas of the posterior cortex are involved with cognitive task performance.
For math tasks, it appears that diffi-culty-free high performance is associated with low left-parietal and right-temporal alpha amplitude, high alpha frequencies in these areas, and high alpha frequency in the left-temporal area. The parietal alpha amplitude findings are in agreement with several past studies of EEG alpha asymmetry [20, 22, 70] while the frequency findings are in partial agreement with Giannitrapani

[71]. Most notably, the relatively right-temporal amplitude activation finding is specifically tied to good performance and in conjunction with the left-parietal findings, it suggests an explanation of discrepancies in laterality studies that have used only one measure of hemispheric specialization [72, 73].

For the spatial task, performance across all subjects appeared to be unrelated to alpha measures of amplitude and frequency. This finding is reminiscent of Giannitra-pani's [78] observation that EEG relationships with performance IQ are generally low or nonexistent. However, across-subject difficulty with this task appeared to be associated with high-temporal lobe alpha amplitude.

This finding is congruent with the results of Willis et al.

[74], who found that in general, high-spatial-ability subjects had greater temporal- than parietal-lobe amplitude activation than low-ability subjects. The finding, in addi-tion, has been replicated, at least for the left-temporal region, in an examination of subjective performance-related experience, EEG alpha activity and mental rotation of imagined block figures [75]. The results, however, seem at odds with the within-subject difficulty data that suggested that parietal lobe activation (frequency) was more affected than the temporal region by spatial task demands.

For the verbal condition, correlations between EEG alpha measures performance and difficulty were generally marginal and difficulty-level-specific.

There was

some suggestion, however, that amplitude activation of the parietal lobe and high right-temporal alpha frequency interfered with performance. These findings are not supportive of BEG asymmetry studies focusing on digit recall [76] or word recognition [21, 77] in which it was reported that relative left-hemisphere alpha amplitude activation predicted high performance. They are, however, similar to the findings of Giannitrapani [78], who reported bilateral involvement of parietal regions and right posttemporal regions within the alpha band for active verbal tasks requiring the use of memory.

Confusion

A sense of confusion experienced by subjects in the present study may have derived from several sources including interference of stimulus items with ongoing operations, similarities between stimuli and representations in short-term memory, low self-esteem and loss of vigilance, and unfamiliarity with task demands. Nev-ertheless, with the exception of the easiest verbal task, high confusion was associated with high alpha amplitude and low alpha frequency recorded from various electrode sites across the 3 different tasks. High right-tempo-ral alpha was found for confused subjects in the math condition while high amplitude in both temporal regions was associated with spatial task confusion. One possible interpretation of these results assumes that interference is a source of this confusion and that its primary effect is

to reduce information processing and thereby induce high alpha amplitude in task or functionally specific regions. If this is the case, then confusion and performance might be correlated and have similar EEG corre-lates. In the math condition, this apparently was the case, while in spatial condition, no reliable interrelationships were found. However, both spatial task confusion and subjective difficulty were associated with high tem-poral-lobe alpha and were highly correlated with one another.

Other EEG alpha correlates to confusion included associations with low-parietal frequency and amplitude activation, but these tended to emerge only at higher levels of difficulty. Perhaps the vulnerability of the temporal lobes to the effects of interference is related to its

functional role in the selection of cues in visual discrimination tasks [48] while parietal-lobe involvement in short-term memory [67] underlies the difficult task asso-ciations.

What of the odd finding that confusion was negatively associated with parietal-amplitude activation during the easiest verbal task but positively related to parietal alpha during the most difficult task? A reasonable interpretation is that when cognitive demand is low for verbal (nonvisual) tasks, little alpha blocking or cortical activation occurs, but as demand increases such that automa-tized cognitive routines cannot be called upon, information processing rate increases and activation increases accordingly. Low ability, performance, or high confu-sion, according to this notion, should be associated with high activation levels at low apriori determined difficulty because of this increased demand relative to high-ability subjects.

At high levels of apriori determined difficulty, how-ever, low-ability subjects would suffer relative to subjects of high ability a greater decrease in information processing rate due to increasing interference and confusion and hence show lower levels of activation. This proposition is supported by lateral eye movement research in this laboratory which found that among verbal-izers, relatively right hemispheric activation was associated with better verbal task performance during a stress-free condition while among all subjects in a stress condi-tion, performance was negatively related to relatively right-hemispheric activation [63].

Anxiety

The relationship between EEG alpha activity and the experience of anxiety is still obscure even though it has been a central concern of clinical investigations as far

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back as the 1930s [79]. In anxiety neurotics and normals subjected to stress, reduced alpha abundance and acceleration of alpha frequencies have been frequently reported [25-28]. However, slow-wave activity in the form of delta, theta and slow alpha activity has been observed in anxiety states as well [80, 81]. In the present study, performance anxiety was found to increase across subjects as a function of either relatively right-parietal or left-temporal activation. Specific correlations were affected by type of task and task difficulty. In 2 cases, state anxiety was directly related to high right-temporal (spa-tial condition) and low right-parietal (easiest verbal task) alpha amplitude but as discussed earlier, most typically anxiety correlated with the ratio measure only. Math trait anxiety was associated with frequency measures recorded during the less difficult tasks from the left-pari-etal and righ-temporal leads. Taken collectively these data make it very clear that attempts to identify the relationship of anxiety with alpha when using a single electrode placement [82, 83], attempts to localize cortical components of anxiety to one hemisphere or the other [84, 85] and efforts to explicate the physiological correlates of anxiety without consideration of cognitive components are inadequate and potentially misleading.

The observed across-subject relationships between

EEG alpha measures and anxiety are congruent with the results of some studies which have directly manipulated affect and which have indicated that relatively right-parietal [23, 86] and left-temporal (T;) [84] amplitude reductions occur during negative affective states. How-ever, other investigations, some of which have specifically focused on anxiety, have reported no laterality effects for parietal and temporal sites [87,

88], and

greater relative left-parietal activation compared to baseline values [85] and bilateral activation in parietal regions [89]. In addition, Davidson and co-workers [23, 86, 90] have consistently reported that valence of emo-tion, while associated with frontal alpha asymmetry, is not related to parietal asymmetries.

That the perceived source of anxiety or type of fear is an important variable in determining associated patterns of hemispheric asymmetry is supported by both clinical and experimental studies [91-95]. For example, left-temporal lobe epileptics tend to be more self critical and have greater fear of social and sexual situations than right temporal lobe epileptics even though the two groups exhibit no differences in overall reported fear [95]. Studies which have specifically investigated performance anxiety or social stress in normals have generally used lateral eye movements as a dependent measure.

Most of these studies have reported either right-hemi-spheric activation in association with stress or embarrassment [29, 31, 63] or inhibited right hemispheric functioning in trait-anxious subjects [63, 96]. Given the reported relationship between direction of lateral eye movement and parietal alpha amplitude asymmetry [66], the results of the present study are in full agreement with the lateral eye movement findings. Finally, while the results of Davidson's group indicate no correlation between parietal alpha asymmetry and intensity of negative affect, their research has generally focused on sadness and feelings of depression. In a recent investigation of fear and happiness, Ahern and Schwartz [97] have found that relatively right-parietal activation does predict intensity of affect.

Diverting of Attention

The present investigation revealed a robust association between the degree to which subjects diverted their attention away from the task and right-parietal amplitude activation. This relationship was found for all tasks but appears to be specific to stimulus-driven conditions, since the relationship is not found for self-generated tasks [75]. While it is possible that subjects exhibited more alpha blocking due to longer periods of visual fixation to the extent that they diverted their attention away from the auditory test stimuli, the unique correlation with right-parietal activation remains unexplained.

Recent research into the functions of the parietal lobe indicated that it plays a significant role in the control of directed attention in nonhuman primates [98, 99]. In humans, the control of attention appears to be lateral-ized to the right-parietal region [100-102]. More specifi-cally, Posner et al. [103] have provided experimental evidence that an attention disengagement mechanism may be localized in the right-parietal region. However, in the absence of external stimulation, when subjects are required to be vigilant, for example, during self-generated tasks, there is evidence implicating involvement of the left hemisphere [104-106]. Furthermore, under conditions of high arousal, left-hemispheric dominance and narrowly focused rigid style of attention deployment may emerge which may be both a consequence of that level of arousal and an attempt to reduce associated feelings of anxiety [30, 96]. The results of the present study appear to be in agreement with this conceptualization of the control of atten-tion. The switching of attention was highly correlated with anxiety, but was not generally associated with left-hemispheric activation. Only in the most anxiety-induc-ing task, the most difficult math task, was this attentional shifting tendency reliably associated with left-tem-poral lobe activation.

Guessing Strategy

Reliance on the use of a guessing strategy was not directly manipulated in this study, but across-subject correlational analyses did yield some significant data with regard to alpha activity. The most reliable correlations were found with relatively left-temporal lobe amplitude activation during the math condition (easy task) and relatively left-parietal frequency activation during the verbal condition (intermediate-difficulty task). Since reduced amplitude

activation in the right-temporal area was associated with math confusion and relatively right-temporal activation with performance, the correlate with guessing may merely reflect what low-performing subjects must do given an opposite pattern of tonic activa-tion. A similar argument might be made for the verbal tasks observation since low-performing subjects exhibited a trend towards guessing as a function of low right-parietal frequency activation. But as discussed previous-ly, guessing may affect alpha asymmetry by increasing information processing and decreasing the effects of high difficulty, confusion and anxiety. Future research might directly manipulate these variables in order to tease out tonic and phasic arousal effects.

Conclusions

Thirty years ago, the prevailing view among elec-troencephalographers was that 'gross' EEG and its component waveforms were far too primitive in nature to reflect the utilization of higher cognitive processes [107].

Today, thanks to the advent of the computer, the use of more sophisticated waveform analysis and statistical techniques and a wedding between neuropsychology and psychophysiology, this view is no longer tenable. But the

*science of squiggly lines' in a vary basic sense still remains a correlative endeavor seeking hypothesis to fit data rather than testing hypothesis based on data. In this report, as in others, after the fact analysis prevails, perhaps unavoidably.

Post hoc analysis aside, the results of this investigation suggest tentative conclusions concerning the dependent measures of EEG alpha asymmetry may be made.

First, what are the similarities and differences between measures of 'activation', alpha frequency and alpha am-plitude? The results of the present report suggest that while these measures are often correlated, they are sensitive to different cognitive dimensions. Secondly, what is the relative importance of performance anxiety and more strictly cognitive factors such as strategy in determining patterns of alpha asymmetry? The current study suggests that emotion and emotion-related cognitive factors (e.g. diverting of attention) may play a larger part in effecting alpha asymmetry than previously thought. Thirdly, what are the precise relationships between alpha activity parameters and cognitive task performance and performance related variables? While data in the current investigation generally supports the notion that greater alpha activation is associated with greater performance, difficulty and anxiety and verbal task results suggest this proposition is only provisionally valid. Task type, difficulty level and electrode position appear to be moderating variables. Finally, while the study clearly identifies difficulty as an important factor to consider in EEG investigations of hemispheric specialization, the relative importance of difficulty induced changes in cognitive effort, strategy and emotionality remain to be deter-mined.

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